

DSA Day 42: Searching & Hash Functions

Book: Data Structures Through C in Depth - Ch9 §9.1-9.3.1 · pp 472-479 | Code in Python

Overview

Linear search $O(n)$; binary search $O(\log n)$ on sorted data. Hashing maps keys to buckets for average $O(1)$ lookup.

Key concepts to master

- binary search
- hash function goals
- truncation, mid-square, folding

Python implementation

```
def binary_search(a, x):
    lo, hi = 0, len(a) - 1
    while lo <= hi:
        m = (lo + hi) // 2
        if a[m] == x: return m
        if a[m] < x: lo = m + 1
        else: hi = m - 1
    return -1
```

Complexity

Binary search $O(\log n)$; hash avg $O(1)$.

Common pitfalls

- Binary search on unsorted data
- Poor hash causing clustering

Practice

- Binary-search a sorted array
- What makes a good hash function?

